Hall Ticket Number:

MR-22

Code: B325CB2S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Discrete Mathematics

(Common to CSB and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

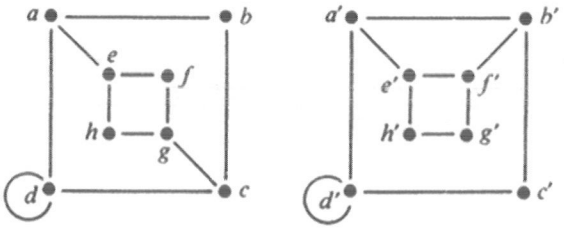
Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define Tautology.	1	1	1	1
1. b)	Find (P → (P V Q)) formula is tautology or contradiction.	1	1	1	1
1. c)	Define Venn Diagram with Example.	1	1	2	1
1. d)	What is Hasse Diagram?	1	1	2	1
1. e)	Define Group.	1	1	3	1
1. f)	Define Lattice.	1	1	3	1
1. g)	If P(n,4) = 42 P(n,2) (permutation),find the value of n?	1	2	4	1
1. h)	How many persons must be chosen in order that at least five of them will have birthdays in the same calendar month?	1	2	4	1
1. i)	Define bipartite graph.	1	1	5	1
1. j)	What is Chromatic number?	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that (¬P Λ (¬Q Λ R)) V (Q Λ R) V (P Λ R) ⇔ R.	5	3	1	1
2. b)	Define Free and Bound variables, Predicates with example.	5	1	1	1
	OR				
2. c)	Obtain PDNF of (Q→P) Λ (¬P Λ Q).	5	3	1	1
2. d)	Show that R can be derived from the premises P → Q, Q → R and P.	5	2	1	1
3. a)	Find the inverse of the function $f(x) = x^2+2$.	5	1	2	1
3. b)	Compute the adjacency matrix of the transitive closure on the set G={a,b,c,d,e},and R={a,b),(b,c),(c,d),(d,e),(e,d)}.	5	2	2	1
	OR				
3. c)	For A={1,2,3,4}, Let R={(1,1),(1,2),(2,3),(3,3),(3,4)} be a relation on A. Find R^2 R^3 R^4 and draw their diagram.	5	2	2	1
3. d)	Let R be a relation on the se t of all ordered pairs of natural numbers defined by (x,y) R (u,v) iff xv=yu. Show that R is an equivalence relation.	5	3	2	1
4. a)	Let G={1,-1,i,-i}, Judge whether G forms an abelian group under usual multiplication.	5	2	3	1
4. b)	Write the properties of algebraic system with an example.	5	1	3	1
	OR				
4. c)	In any group (G,*), by proving the inverse of every element is unique. Show that $(a*b)^{-1} = b^{-1} * a^{-1}$, for all a,b ∈ G.	5	3	3	1
4. d)	Let * be a binary operation on the set R of real numbers defined by a*b=a+b+2ab.Is (R,*) a monoid? Justify.	5	3	3	1
5. a)	How many ways can we get a sum of 8 when two indistinguishable dice are rolled?	5	3	4	1
5. b)	How many different permutations of the letters are there in the word "COMPUTERS".	5	3	4	1
	OR				

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5. c)	Use Multinomial theorem to Expand $(X_1+X_2+X_3+X_4)^4$.	5	3	4	1
5. d)	How many integers between 1 and 300(inclusive) are divisible by at least one of 5, 6, 8.	5	3	4	1
6. a)	Let G be a 4-regular connected planar graph having 16 edges. Find the number of regions of G.	5	2	5	1
6. b)	Show that the two graphs are not isomorphic.	5	3	5	1
					
	OR				
6. c)	Let D be the digraph whose vertex set is V={v1,v2,v3,v4,v5} and the directed edge set is E={(v1,v4),(v2,v3),(v3,v5),(v4,v2),(v4,v4),(v4,v5),(v5,v1)}Draw a diagram of D and indicate the out-degree and in-degree of all vertices.	5	2	5	1
6. d)	What is a Hamiltonian Cycle? Draw bipartiate graph $K_{3,4}$ and prove that this graph does not have a Hamiltonian Cycle.	5	1	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome